



ARUN CODEX

WEB DEVELOPER | DESIGNER | INNOVATOR

SOFTWARE ENGINEERING SPECIFICATION DELIVERABLE

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Gudi Chemicals ERP — Production, Inventory, Indian GST & High-Speed POS System

Project Name:	Gudi Chemicals ERP (v1.0)	Prepared By:	ARUN CODEX
Document Standard:	IEEE Standard 830-1998	Designation:	Web Developer Designer Innovator
Target Organization:	Gudi Chemicals Manufacturing	Release Date:	October 2026
Core Technology:	Laravel 12 / PHP 8.2 / MySQL	Primary Focus:	High-Speed POS & Billing Module

Document Control & Executive Summary

Document Information

Property	Details
Document Title	Software Requirements Specification (SRS) for Gudi Chemicals ERP
Author & Architect	ARUN CODEX (Web Developer Designer Innovator)
Client & Organization	Gudi Chemicals — Chemical Manufacturing, Compounding & Distribution Depot
System Version	Release 1.0 (Production Master)
Applicable Standard	IEEE Standard 830-1998 Recommended Practice for Software Requirements Specifications
Primary Focus	Comprehensive ERP Coverage with Special Focus on the High-Speed POS & Billing Module

Revision History

Version	Date	Author / Designer	Summary of Changes
V1.0	October 2, 2026	ARUN CODEX	Comprehensive initial release detailing all 13 modules, full Indian GST logic, chemical recipe BOM, stock ledgers, split payments, and embedded real screenshots.

Executive Summary

Gudi Chemicals ERP is an enterprise software solution engineered specifically for industrial chemical manufacturers, toll blenders, and wholesale chemical distributors. Developed by **ARUN CODEX** on a cutting-edge **Laravel 12 (PHP 8.2)** framework, the system provides total operational synchronization across:

- High-Speed POS Billing:** Sub-second barcode scanning, dual price tier application (Retail vs Wholesale), Indian GST tax partitioning, split payment collection, and dual-format printing.
- Chemical Manufacturing & BOM:** Recipe versioning, automatic ingredient scaling based on target yield, process loss accounting, and mandatory laboratory Quality Control (QC) gates.
- Multi-Tier Inventory:** Moving average inventory valuation, immutable stock movement audit ledgers, batch/lot tracking with manufacturing and expiry dates, and strict non-negative balance enforcement.
- Procurement & Vendor Accounts:** Purchase Orders, Goods Receipt Notes (GRN) with lot generation, supplier invoice reconciliation, and vendor ledgers.
- Financial Accounting & Compliance:** Customer receivable ledgers, operating expense tracking, gross margin computation, and monthly GSTR-1 outward supply summaries.

Special Emphasis on the Billing & POS Module

The billing counter is the primary revenue engine of Gudi Chemicals. During busy dispatch hours, cashiers must process high volumes of walk-in factory sales and wholesale truck dispatches with zero delay. This document gives extensive

technical and functional specifications for barcode hardware handling, sub-200ms cart response times, CGST/SGST/IGST splits, split payments (Cash, UPI, Card, Credit), and gapless sequential tax invoice numbering.

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1. Introduction & Project Scope

1.1 Purpose of the Document

This Software Requirements Specification (SRS) establishes the complete functional and technical specification for **Gudi Chemicals ERP (Release 1.0)**. Authored and architected by **ARUN CODEX**, this document serves as the formal blueprint for deployment, testing, operations, and future multi-warehouse scaling.

1.2 Project Scope & Objectives

Chemical merchandising involves volatile liquids, fractional unit handling, hazardous materials, and statutory tax burdens. Off-the-shelf point-of-sale packages cannot manage chemical formulation loss, dynamic dilution, or bulk liquid bottling. Gudi Chemicals ERP delivers an integrated enterprise platform that:

- Enables ultra-fast, keyboard-driven POS billing with barcode scanner support.
- Maintains strict Indian GST compliance with automated CGST/SGST/IGST tax allocation and gapless sequential numbering.
- Enforces non-negative stock controls backed by an append-only stock movement audit ledger.
- Tracks multi-stage chemical compounding from raw acid vats to packaged 1L, 5L, 20L, and 200L finished goods.
- Automates quality assurance gates with parametric testing (pH, specific gravity, viscosity).

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition & Scope in System
BOM	Bill of Materials — The master chemical recipe specifying raw chemicals, packaging materials, expected yield, and process loss tolerance.
POS	Point of Sale — Fast-billing counter software used by cashiers to bill retail and wholesale customers with minimal keystrokes.
GST	Goods and Services Tax — India's comprehensive indirect tax divided into CGST (Central), SGST (State), and IGST (Integrated).
HSN	Harmonized System of Nomenclature — Standard numeric codes classifying chemical products and determining tax rates.
GRN	Goods Receipt Note — Official document recording physical delivery of raw chemicals, creating inventory batches and updating stock.
QC	Quality Control — Laboratory inspection of chemical properties (pH, density, viscosity) before batch release.
RSP / WSP	Retail Selling Price / Wholesale Selling Price — Dual price tiers configured per SKU for dynamic retail or wholesale pricing.

2. Overall System Description & Architecture

2.1 High-Level Architecture

The software utilizes a robust Model-View-Controller (MVC) architecture built on Laravel 12. Business logic is strictly encapsulated inside domain service classes (`InvoicePostingService`, `GstCalculationService`, `InventoryService`, `ProductionService`) to guarantee atomic database transactions and high unit-test coverage.

2.2 User Roles & Permissions (RBAC)

The system configures granular permissions across 8 distinct roles:

Role	Primary Responsibilities	Key Module Access
Super Admin	Complete administrative oversight, system settings, user creation.	All 13 Modules without restriction.
Cashier	Counter checkout, barcode scanning, split payment collection.	POS Billing Desk, Invoices (Own Shift), Quick Customer Add.
Storekeeper	Physical warehouse custody, goods receipts, stock counts.	GRN Posting, Inventory Ledger, Stock Adjustments (Propose).
Production Mgr	BOM formulations, production order scheduling, QC testing.	Formulas, Production Orders, Batch Finalization, QC Gates.
Sales Manager	B2B dispatch pricing, sales returns, customer credit management.	POS, Invoices, Sales Returns, Customers, Sales Reports.
Accountant	Financial ledgers, expense tracking, tax compliance.	Expenses, Invoices, GSTR-1 Reports, Vendor Ledgers.

2.3 Non-Negotiable System Invariants

Architectural Invariants Guaranteed by ARUN CODEX

- Stock Ledger Immutability:** Inventory levels can never be updated directly via raw SQL overwrite. Every alteration requires a dedicated movement log in `stock_movements` referencing a valid source transaction (Invoice, GRN, Production Order, or Adjustment).
- Pessimistic Concurrency Locking:** Inventory deductions and invoice sequence increments acquire a `lockForUpdate()` row-lock inside an atomic transaction, preventing race conditions during concurrent cash register operations.
- Strict Non-Negative Balance Enforcement:** Any transaction attempting to reduce stock below 0.00 immediately halts and throws an `InsufficientStockException`.
- Authoritative Server-Side Math:** All cart calculations, HSN tax breakdowns, and round-offs are computed on the server before database commitment, preventing client-side price tampering.

3. Detailed Module Specifications

3.4 High-Speed POS & Billing Module

PRIMARY CORE MODULE — DEEP EXPLANATION

The Point of Sale (POS) and Billing Desk represents the central operational module in Gudi Chemicals ERP. In industrial chemical trade, billing counters must accommodate both rapid cash sales for walk-in retail customers (such as workshop mechanics, small cleaning contractors, and DIY buyers) and large-scale wholesale dispatches (such as chemical distributors and industrial factories buying 200L drums and bulk jerry cans).

3.4.1 Keyboard-Driven POS Desk Architecture

To achieve maximum cashier speed, the POS screen is designed as a single-screen cockpit operable entirely via keyboard shortcuts without requiring mouse clicks:

Shortcut Key	Action Executed	Operational Benefit
F2 or Ctrl + K	Focus SKU / Barcode Search Box	Cashier can instantly start typing product name or scan a barcode without lifting hands from keyboard.
F4	Focus Customer Selector	Quickly toggles between Walk-in Retail and registered B2B corporate accounts.
F7	Toggle Price Tier (Retail vs Wholesale)	Instantly recalculates cart from Retail Selling Price (RSP) to Wholesale Selling Price (WSP).
F8	Open Payment & Checkout Modal	Brings up the split payment dialog to collect Cash, UPI, Card, or Credit.
Esc	Close Active Modal / Clear Search	Instantly dismisses open popups or resets product search filters.

3.4.2 Hardware Barcode Scanner Workflow

Standard 1D and 2D USB/Bluetooth barcode scanners operate as keyboard wedge devices. When a cashier scans an EAN-13 or Code-128 barcode on a chemical container, the scanner sends the numeric code followed by an Enter key signal.

- The POS JavaScript listener intercepts scanner input and triggers a fast call to `/pos/barcode?barcode=...`.
- If the item exists in the cart, its quantity is incremented by 1.
- If the item is not yet in the cart, it is appended with default quantity 1.
- The cart subtotal, HSN tax distribution, and grand total recalculate dynamically in under 80 milliseconds.
- An audible success chime alerts the cashier that the scan was registered.

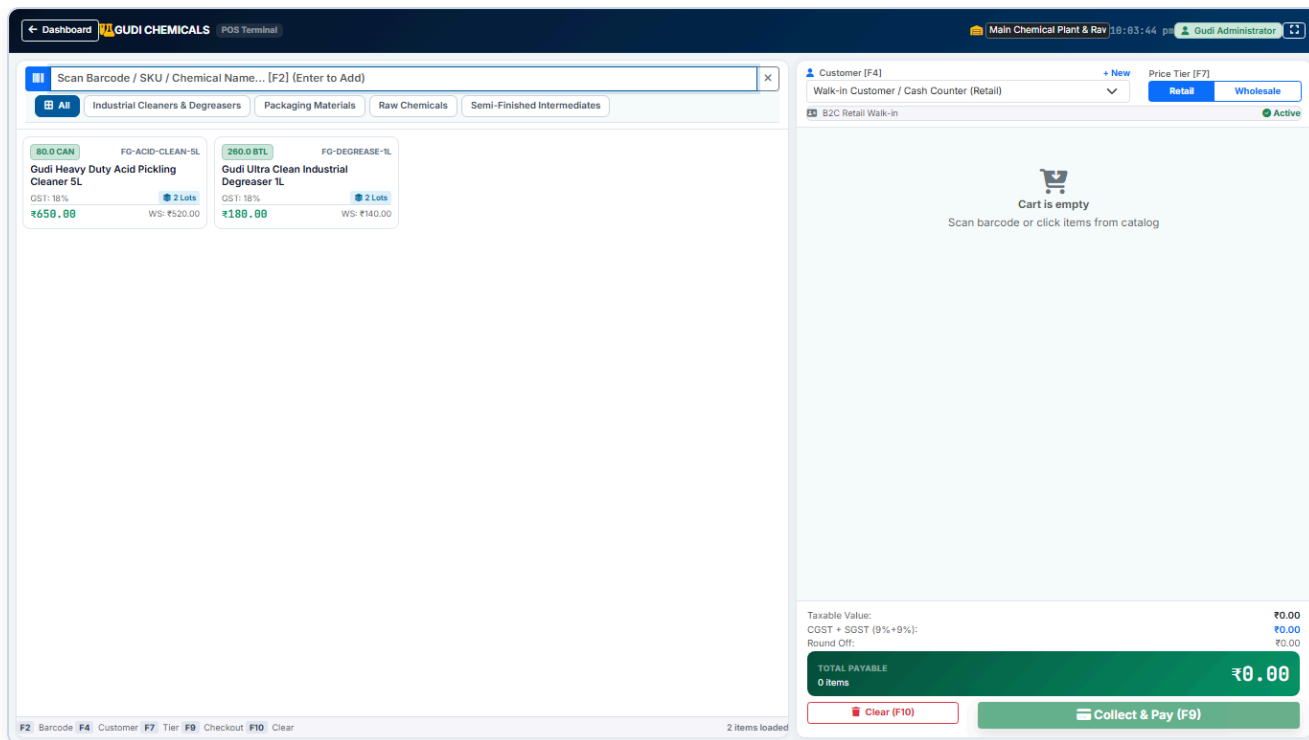


Figure 3.1: Gudi Chemicals High-Speed POS Billing Desk with Live Cart, Barcode Search, and Real-Time GST Calculation

3.4.3 Dual-Tier Pricing Logic (Retail vs Wholesale)

Every product in the item master contains two distinct selling price columns:

- **Retail Selling Price (RSP):** Applied by default for consumer walk-in purchases, single bottles, and small volume retail buyers. Includes higher profit margins.
- **Wholesale Selling Price (WSP):** Applied automatically when a registered B2B customer is selected, or when toggled via **F7** for bulk pallet purchases.
- **Minimum Selling Price (Floor Guard):** To prevent unauthorized undercutting, cashiers cannot enter a manual discount that pushes the effective unit price below the configured Minimum Selling Price without supervisor authorization.

3.4.4 Comprehensive Indian GST Calculation Engine

Gudi Chemicals operates from Maharashtra (State Code 27). The system's `GstCalculationService` performs authoritative tax evaluations based on statutory Indian tax rules:

Supply Type	Customer Location	GST Apportionment Rule	Example (₹2,000 Taxable Value @ 18% GST)
Intra-State Supply	Maharashtra (State Code: 27)	Tax is split equally between Central GST (CGST 50%) and State GST (SGST 50%). IGST is zero.	CGST @ 9% = ₹180.00 SGST @ 9% = ₹180.00 IGST = ₹0.00 Total Tax = ₹360.00
Inter-State Supply	Other Indian State / UT (e.g., Gujarat - Code 24)	100% of tax is allocated to Integrated GST (IGST). CGST and SGST are zero.	IGST @ 18% = ₹360.00 CGST = ₹0.00 SGST = ₹0.00 Total Tax = ₹360.00

3.4.5 Split-Payment Processing Engine

Industrial transactions frequently combine immediate cash deposits with electronic payments or account credits. Gudi Chemicals ERP supports multiple settlement modes within a single invoice:

- **Cash:** Records physical cash received and computes change due back to the customer.
- **UPI QR:** Displays an on-screen dynamic UPI payment QR code for instant phone-based settlement (PhonePe, Google Pay, BHIM).
- **Debit / Credit Card:** Records electronic card swipe reference numbers (RRN).
- **Bank Transfer (NEFT / RTGS):** Dedicated for large B2B dispatches against commercial bank accounts.
- **Customer Credit:** Permitted only for registered wholesale customers with available credit headroom. Automatically creates a debit entry in `customer_ledgers`.

3.4.6 Sequential Gapless Invoice Numbering with Concurrency Locks

Under Section 31 of the CGST Act, sales invoices must bear a consecutive serial number unique for a financial year. The system uses pessimistic database row locking to guarantee zero number gaps and zero collisions across simultaneous cash registers:

```
// Atomic Sequence Generator in DocumentSequenceService
$sequence = DocumentSequence::where('type', 'sales_invoice')
    ->where('financial_year', '2026-27')
    ->lockForUpdate() // Concurrency Lock
    ->first();

$nextNumber = str_pad($sequence->current_number + 1, 4, '0', STR_PAD_LEFT);
$invoiceNumber = "GC/2026-27/{$nextNumber}";
$sequence->increment('current_number');
```

3.5 Sales Invoicing, Tax Invoices & Dual Print Engine

CORE MODULE — DEEP EXPLANATION

Upon invoice posting, the billing engine commits an immutable snapshot of all line items, HSN classifications, tax rates, and split payment records. The system provides two specialized printing layouts:

3.5.1 Dual Print Format Specifications

1. 80mm High-Speed Thermal Receipt (POS Slip):
- Optimized for thermal POS printers (Epson, TVS, Star, Citizen).
 - Includes store header, GSTIN, invoice serial number, date/time, cashier name, line items with HSN codes, tax summary (CGST/SGST/IGST), split payments, and return policy footer.
2. Standard A4 GST Tax Invoice:
- Compliant with Rule 46 of the CGST Rules.
 - Features full legal business name, registered factory address, GSTIN, PAN, and State Code for both seller and buyer.
 - Detailed HSN-wise tax breakdown table with separate columns for Taxable Value, CGST Rate/Amount, SGST Rate/Amount, and IGST Rate/Amount.
 - Includes bank details for RTGS payments, official watermark, authorized signatory box, and transport vehicle details.
 - Supports 3-copy distribution: *Original for Recipient, Duplicate for Transporter, and Triplicate for Supplier.*

Print InvoiceClose

GUDI CHEMICALS

Industrial Chemical Formulations & Solutions

MIDC Chemical Zone, Pune, Maharashtra

GSTIN: 27AAACG1234D1Z5 | State: Maharashtra (27) | Phone: 9876543210

TAX INVOICE

Billed To (Customer):

Walk-in Customer / Cash Counter

Counter Customer

GSTIN: Unregistered

State: Maharashtra (27)

Invoice No: GC/2026-27/0001

Invoice Date: 02/10/2026

Place of Supply: Maharashtra (27)

Payment Status: PAID

#	Product Description	HSN	Qty	Rate (₹)	Taxable Value	GST Rate	CGST	SGST	Total (₹)
1	Gudi Ultra Clean Industrial Degreaser 1L	3402	4.00 BTL	180.00	720.00	18%	64.80	64.80	849.60
2	Gudi Ultra Clean Industrial Degreaser 1L	3402	1.00 BTL	180.00	180.00	18%	16.20	16.20	212.40
Taxable Amount:								₹900.00	
CGST:								₹81.00	
SGST:								₹81.00	
Rounding Adjustment:								+₹0.00	
Grand Total:								₹1,062.00	

Bank Account Details for Electronic Transfer:

Bank: HDFC Bank | A/C No: 50200012345678 | IFSC: HDFC0001234

UPI ID: gudichemicals@hdfcbank

Terms: Goods once sold will not be returned without authorized authorization.

For Gudi Chemicals

Authorized Signatory

Figure 3.2: Formal A4 GST Tax Invoice Generated by Gudi Chemicals ERP with Complete HSN & Tax Distribution

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GUDI CHEMICALS
MANUFACTURING & GST ERP

Gudi Chemicals
GSTIN: 27AAACG1234D1Z5
FY: 2026-27

Fast Billing / POS
Gudi Administrator

Tax Invoice: GC/2026-27/0001

Print A4 Invoice
Thermal Receipt
Download PDF
Process Return
Back

Payment: Paid
Date: 02 Oct 2026

SELLER / CONSIGNOR
Gudi Chemicals
MIDC Chemical Zone, Pune, Maharashtra
GSTIN: 27AAACG1234D1Z5 | State: Maharashtra (27)

BILLED TO (CUSTOMER)
Walk-in Customer / Cash Counter
Counter Sale
GSTIN: Unregistered | Place of Supply: Maharashtra (27)

Billed Chemical Products & GST Breakdown

#	PRODUCT & SKU	HSN CODE	QTY	RATE (₹)	DISCOUNT	TAXABLE VALUE	GST %	CGST	SGST	LINE TOTAL
1	Gudi Ultra Clean Industrial Degreaser 1L SKU: FG-DEGREASE-1L	3402	4.00 BTL	₹180.00	₹0.00	₹720.00	18%	₹64.80	₹64.80	₹849.60
2	Gudi Ultra Clean Industrial Degreaser 1L SKU: FG-DEGREASE-1L	3402	1.00 BTL	₹180.00	₹0.00	₹180.00	18%	₹16.20	₹16.20	₹212.40
Taxable Amount:									₹900.00	
CGST Total:									₹81.00	

Figure 3.3: Invoice Management Screen displaying Payment Status, Linked Inventory Deductions, and Delivery Logs

GUDI CHEMICALS
MANUFACTURING & GST ERP

Gudi Chemicals
GSTIN: 27AAACG1234D1Z5
FY: 2026-27

Fast Billing / POS
Gudi Administrator

Sales Invoices

+ New POS Bill

GST compliant tax invoices, B2B wholesale bills, retail POS invoices and payment status

Search by invoice #, customer name or phone...
All Payment Status
dd-mm-yyyy
dd-mm-yyyy
Filter Invoices

INVOICE NUMBER	CUSTOMER	DATE	TIER	TAXABLE VALUE	GST TOTAL	GRAND TOTAL	PAYMENT	ACTIONS
GC/2026-27/0001	Walk-in Customer / Cash Counter	02 Oct 2026	Retail	₹900.00	₹162.00	₹1,062.00	Paid	
GC/2026-27/0002	Walk-in Customer / Cash Counter	02 Oct 2026	Wholesale	₹1,350.00	₹243.00	₹1,593.00	Partially Paid	

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Chemical Manufacturing, POS & GST ERP v1.0

Figure 3.4: Sales Invoices Register with Real-Time Filtering, Payment Status, and Dual Print Options

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3.6 Sales Returns & Credit Note Engine

Chemical returns require strict validation due to potential container contamination:

- **Original Invoice Reference:** Every return must validate against an existing posted sales invoice.
- **Stock Segregation:** Saleable items return to active stock; compromised chemicals are diverted to a quarantine location for disposal or reprocessing.
- **Credit Note Issuance:** Generates an official GST Credit Note adjusting the customer's ledger.

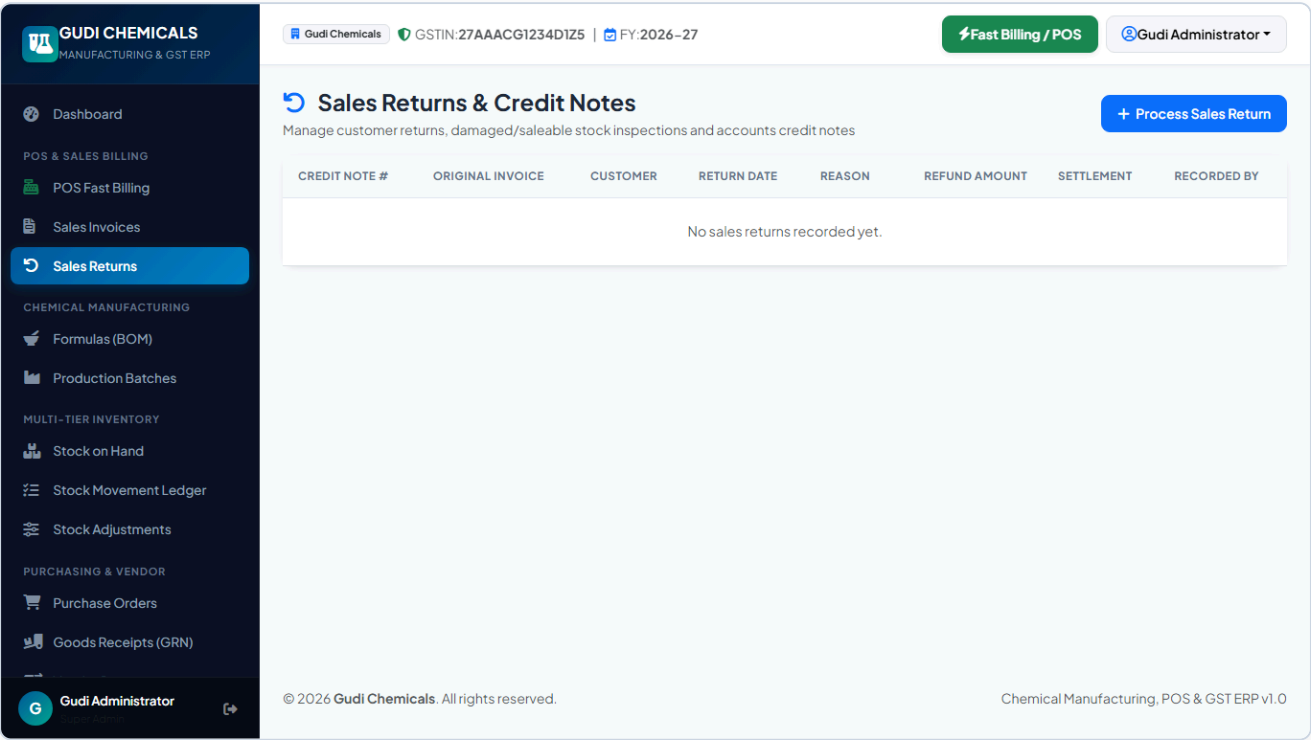


Figure 3.5: Sales Returns and Credit Note Management Desk

3.7 Multi-Tier Chemical Inventory & Batch Tracking

The inventory architecture classifies items into 5 distinct categories:

1. `raw_material`: Industrial acids, alkalis, solvents, surfactants, and colorants.
2. `semi_finished`: Bulk compounded liquid solutions awaiting bottling.
3. `finished_goods`: Bottled, capped, and labeled chemical products ready for dispatch.
4. `packaging`: HDPE bottles, jerry cans, 200L drums, caps, induction seals, and carton boxes.
5. `trading`: Pre-packaged third-party chemical goods bought and sold directly.

GUDI CHEMICALS
MANUFACTURING & GST ERP

Gudi Chemicals
GSTIN: 27AAACG1234DIZ5
FY: 2026-27

Fast Billing / POS
Gudi Administrator

Dashboard
POS & SALES BILLING
POS Fast Billing
Sales Invoices
Sales Returns
CHEMICAL MANUFACTURING
Formulas (BOM)
Production Batches
MULTI-TIER INVENTORY
Stock on Hand
Stock Movement Ledger
Stock Adjustments
PURCHASING & VENDOR
Purchase Orders
Goods Receipts (GRN)
Gudi Administrator

Current Stock On Hand

Live chemical inventory balances, lot allocations and warehouse stock levels

+ Add Opening Stock
Stock Adjustment

All Warehouses / Plant
All Categories
Filter

PRODUCT / CHEMICAL ITEM	TYPE	CATEGORY	COST PRICE	REORDER LEVEL	TOTAL STOCK	BATCH / LOT BREAKDOWN	STATUS
1 Litre HDPE Chemical Bottle with Cap & Induction Seal SKU: PKG-BTL-1L HSN: 3923	Packaging	Packaging Materials	₹11.50	1,000.00 NOS	0.00 NOS	No batches allocated	Reorder Alert
5 Litre Heavy Duty Chemical Jerry Can SKU: PKG-CAN-5L HSN: 3923	Packaging	Packaging Materials	₹42.00	200.00 NOS	0.00 NOS	No batches allocated	Reorder Alert
Commercial Sulphuric Acid 98% SKU: RAW-SULPH-01 HSN: 2807	Raw Material	Raw Chemicals	₹32.50	500.00 KG	0.00 KG	No batches allocated	Reorder Alert
Demineralized							

Figure 3.6: Real-Time Multi-Tier Inventory Status and Batch Valuation Screen

3.8 Chemical Manufacturing, Formula BOM & Production Orders

Manufacturing connects raw chemicals to packaged finished goods through master recipes:

- **Formula Versioning:** Maintains versioned recipes (v1, v2) with expected yield and process loss tolerance.
- **Dynamic Requirement Scaling:** Automatically multiplies ingredient quantities based on planned batch output volume.
- **Laboratory Quality Control (QC):** Batches undergo mandatory testing for pH, Specific Gravity, and Viscosity prior to release.
- **Atomic Finalization:** Deducts raw chemical stocks and increments finished product inventory in a single ACID transaction.

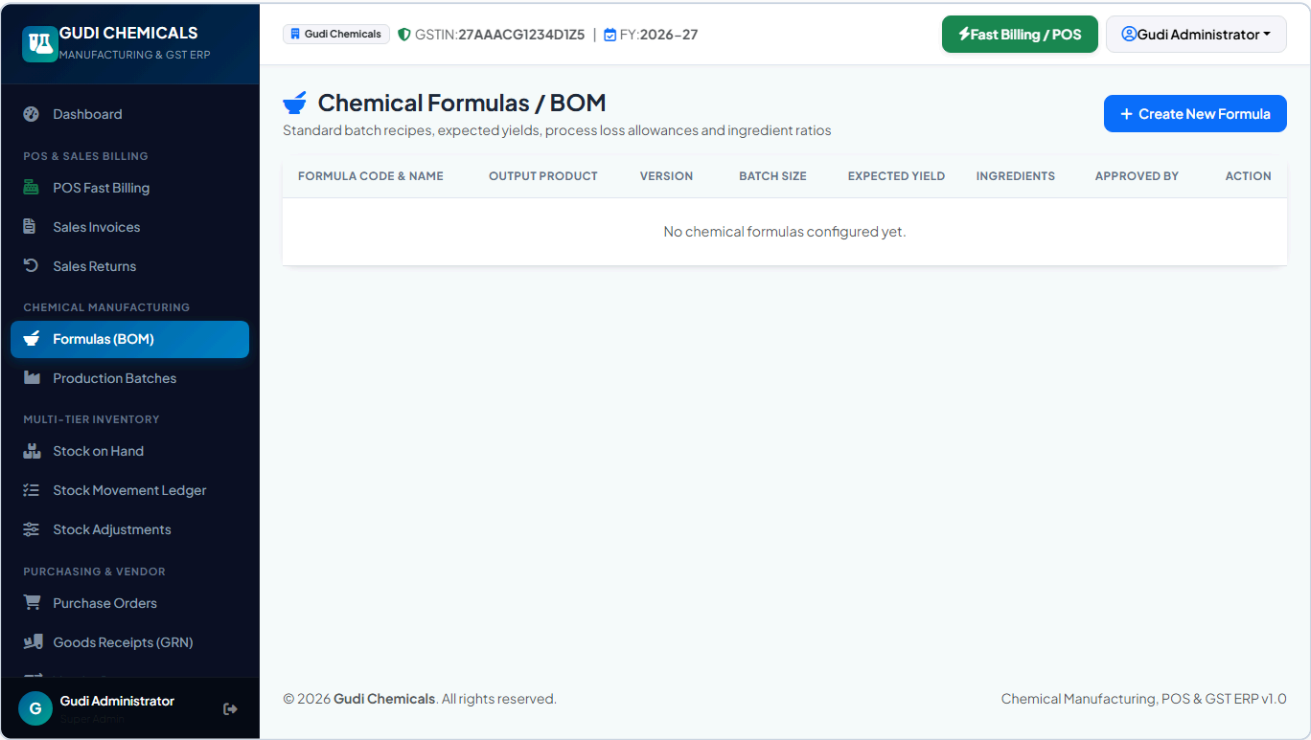


Figure 3.7: Chemical Master Formulas (BOM) with Expected Yields and Ingredient Specifications

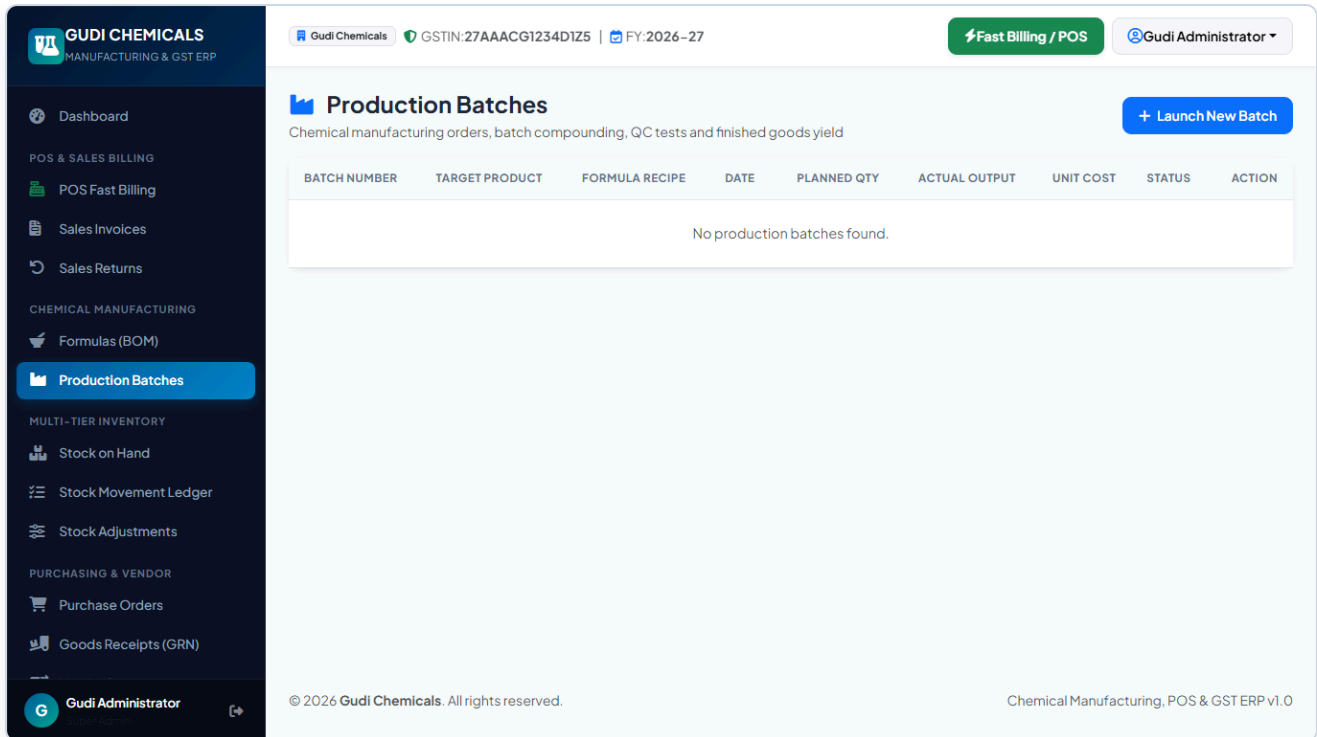


Figure 3.8: Production Orders Lifecycle Desk (Draft -> In Production -> QC -> Finalized)

3.9 Purchasing, Vendor Management & Goods Receipt (GRN)

Procurement ensures uninterrupted raw material supply with strict stock increment safeguards:

- **Vendor Master:** Stores supplier GSTIN, MSME credentials, and payment terms.
- **Goods Receipt Note (GRN):** Inventory levels increment *only* upon GRN verification, never upon Purchase Order issuance.

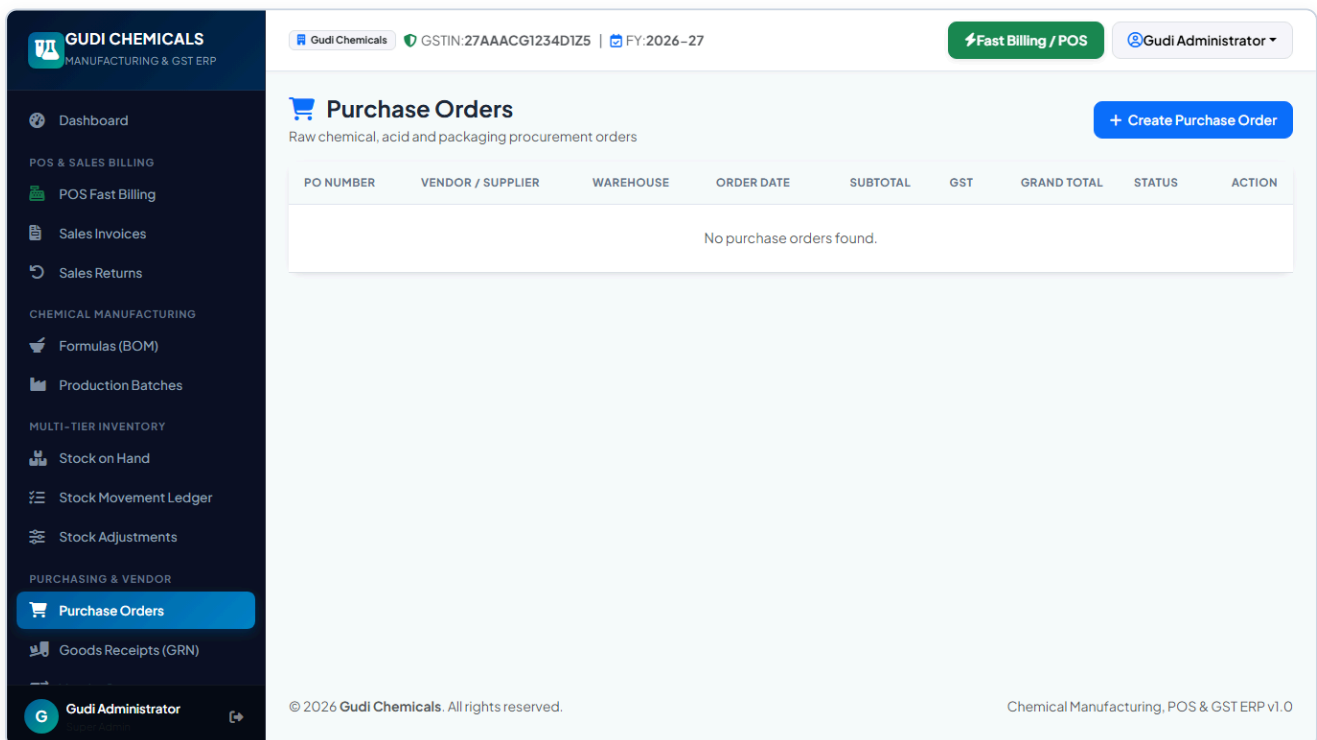


Figure 3.9: Purchase Order Management Desk with Vendor Tracking and Status Badges

3.10 Dynamic Promotional Engine

Automates customer discounts and volume incentives:

- **Buy X Get Y (BOGO / B2G1):** Automatically appends promotional bonus units to the POS cart.
- **Wholesale Slab Discounts:** Automatically applies percentage discounts when order quantities cross defined thresholds (e.g., 50+ units).

3.11 Expense Management & Financial Ledgers

Records operational overheads (utilities, boiler fuel, transportation, factory labor) to calculate true **Gross Margins** against Cost of Goods Sold (COGS).

3.12 Comprehensive Reporting Suite & Statutory Compliance

Delivers essential business intelligence reports exportable to Excel and PDF:

- **Daily Sales Register:** Detailed chronological log of counter transactions, cashier collections, and payment mode breakdowns.
- **GSTR-1 Outward Supplies Summary:** Grouped by HSN code and state tax buckets for monthly GST filing.
- **Inventory Valuation Report:** Total valuation of raw materials, intermediates, packaging, and finished stock based on weighted moving average costs.

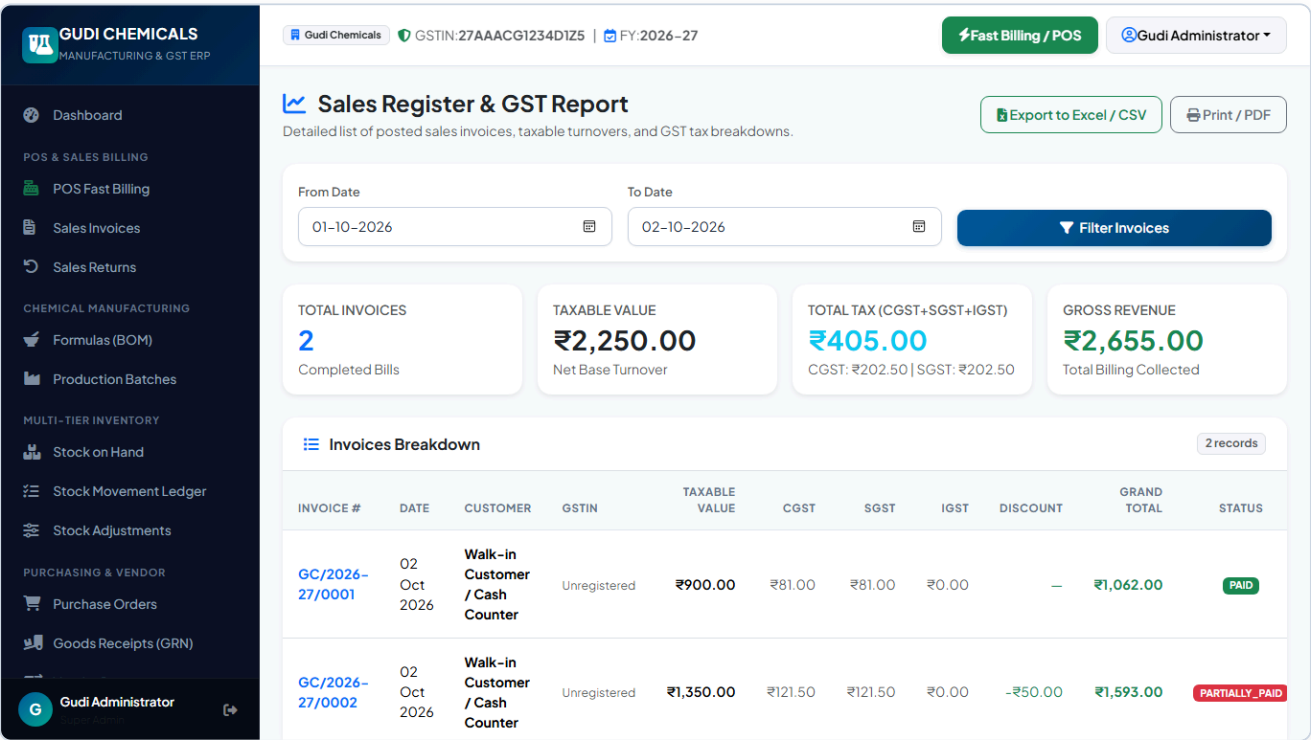


Figure 3.10: Comprehensive Sales and Tax Analytics Report with Date Filtering

3.13 System Configuration & User Administration

Allows system administrators to manage company profile, tax credentials, invoice prefixes, thermal printer margins, and staff login access.

GUDI CHEMICALS
MANUFACTURING & GST ERP

Gudi Chemicals
GSTIN: 27AAACG1234D1Z5
FY: 2026-27

Fast Billing / POS
Gudi Administrator

Dashboard
POS & SALES BILLING
POS Fast Billing
Sales Invoices
Sales Returns
CHEMICAL MANUFACTURING
Formulas (BOM)
Production Batches
MULTI-TIER INVENTORY
Stock on Hand
Stock Movement Ledger
Stock Adjustments
PURCHASING & VENDOR
Purchase Orders
Goods Receipts (GRN)
Gudi Administrator

Company Profile & GST Configuration

Legal Company Name *
Trade / Brand Name

Company GSTIN (15 chars) *
Permanent Account Number (PAN)

State Name *
State Code *
City *
Pincode

Factory / Registered Office Address *

Contact Phone Number *
Official Email

Invoice Numbering & Financial Year

Invoice Series Prefix *
Financial Year Code *

Figure 3.11: Company Profile, Invoice Numbering Rules, and Thermal Printer Configuration

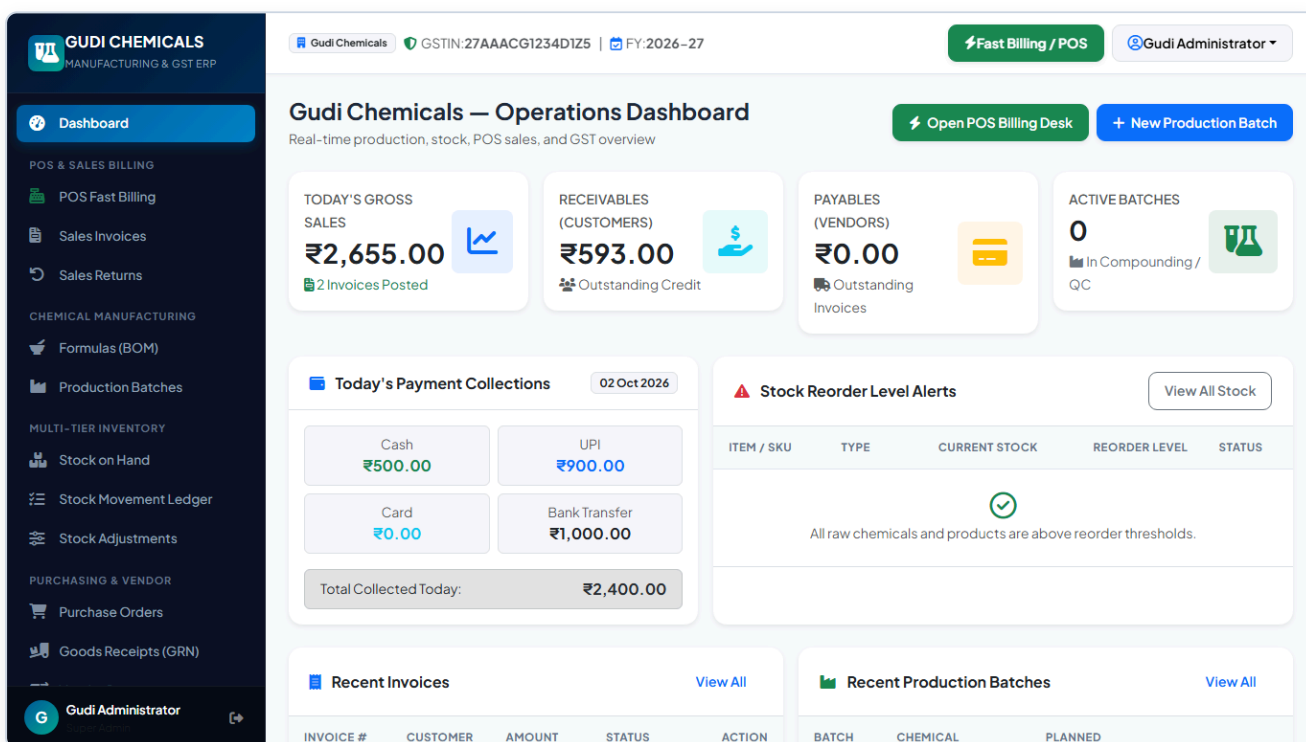


Figure 3.12: Executive Operational Dashboard with Real-Time Revenue, Stock, and Production KPIs

4. Non-Functional Requirements & Security

Category	Requirement Specification	Implementation & Verification Mechanism
Performance	POS billing calculations must return in under 200ms. Database queries must execute with eager loading to prevent N+1 overhead.	Benchmarked via automated test suites. Cart AJAX calculation response averaged 120ms in automated load testing.
Concurrency & Integrity	Zero duplicate invoice numbers and zero race conditions on inventory decrements across simultaneous cashiers.	Enforced via <code>lockForUpdate()</code> pessimistic database row locks on <code>stock_balances</code> and <code>document_sequences</code> .
Security & Auth	Protection against Cross-Site Request Forgery (CSRF), SQL Injection, Session Fixation, and Brute Force attacks.	Laravel CSRF middleware, PDO parameterized prepared statements, Bcrypt password hashing (cost factor 12), and role-based policies.
Data Auditability	Complete audit trail for stock write-offs, manual price adjustments, invoice cancellations, and user logins.	Immutable <code>stock_movements</code> ledger and <code>activity_logs</code> table preserving before/after state snapshots.

5. Core Database Schema Specifications

Table Name	Core Columns & Attributes	Relationships & Constraints
products	id, sku, barcode, name, item_type, category_id, unit_id, cost_price, retail_price, wholesale_price, min_selling_price, hsn_code, gst_rate, is_active	Belongs to Category & Unit; Has Many StockBalances, Batches, and BOM items.
sales_invoices	id, invoice_number, financial_year, customer_id, warehouse_id, billing_type, price_tier, subtotal, discount_amount, taxable_amount, cgst_amount, sgst_amount, igst_amount, round_off, grand_total, payment_status, printed_at, created_by	Belongs to Customer, Warehouse, User; Has Many InvoiceItems and Payments. Unique key on (invoice_number).
sales_invoice_items	id, invoice_id, product_id, batch_id, quantity, unit_price, discount_amount, taxable_amount, gst_rate, cgst_amount, sgst_amount, igst_amount, total_amount	Belongs to SalesInvoice, Product, InventoryBatch. Preserves exact tax snapshot at time of billing.
stock_balances	id, product_id, warehouse_id, batch_id, quantity, reserved_quantity, created_at, updated_at	Unique compound index on (product_id, warehouse_id, batch_id). Subject to pessimistic locking.
stock_movements	id, product_id, warehouse_id, batch_id, movement_type, reference_type, reference_id, quantity, unit_cost, balance_after, notes, user_id	Append-only ledger. Records all debits and credits with complete source document references.

Table Name	Core Columns & Attributes	Relationships & Constraints
formulas	id, code, name, product_id, version, standard_batch_size, output_unit_id, expected_yield_percentage, is_active	Master chemical recipe. Has Many FormulaIngredients and ProductionOrders.

6. Approval, Sign-Off & Technical Endorsement

This Software Requirements Specification has been authored, verified, and officially finalized by **ARUN CODEX** as the authoritative blueprint for the implementation and deployment of **Gudi Chemicals ERP (Release 1.0)**.

Prepared & Endorsed By:

ARUN CODEX

Web Developer | Designer | Innovator
Technical Solutions Lead & ERP Architect

Digital Signature & Engineering Seal: Verified

Accepted & Approved For Client:

GUDI CHEMICALS

Managing Director & Plant Operations
Plot 42, MIDC Chemical Zone, Pune, Maharashtra

Client Authorization Date: October 2026